Table 2. Requirements specification document structure

Overview of System Goals
Operating and Development Environments
External Interfaces and Data Flow
Functional Requirements
Performance Requirements
Exception Handling
Implementation Priorities
Foreseeable Modifications
Design Suggestions

The operating and development environments are sometimes overlooked in requirements specifications. This includes both the browser and the Web server. If you are developing an intranet application, you may be fortunate enough to optimize for a particular browser version. We have found that while a large company may impose a standard browser for the organization for which you've developed an application, another standard may apply to the users in another organization a thousand miles away.

The Web server is perhaps more under your control and certainly less finicky about differences in source code. If you are using PHP, most likely you will be using Apache. It's a good idea to use identical versions of both Apache and PHP for your development and live environments.

For the most part, your list of external interfaces will include the Internet connection between the browser and the Web server, the local file system, and possibly a database connection. We find it helpful to create a diagram that shows the relationship between data elements, the simplest of which might be a box labeled Browser connected to a box labeled Server. The line would have arrows at each end to show that information travels in both directions. This diagram is a description of the context, not a design of the data structure. Whether you will be using a database may be obvious, but which database may not. If the system will be storing data somehow, just show data flowing into a box that could be database or flat file. The goal is to describe how data moves around in the system.

The functional requirements will certainly be the largest part of the document. If you have drawn a data flow diagram, you may have a very good idea of how the system breaks up into modules. The more you can partition the functionality into distinct pieces, the easier it will be to group the functional requirements. We have written